

Electrostatics

The branch of engineering which deals with the static charges is called electrostatics. In static electricity two charged bodies are separated by some insulating medium & the charge remains stationary on the body.

When a particular body is rubbed with another body, electric charge is produced. When a glass rod is rubbed with silk, the glass rod becomes charged.

When two bodies are rubbed together, some amount of charge is transferred from one body to another & this upsets the electrical neutrality of both the bodies. When two bodies are rubbed together, a redistribution of electrons takes place. The body which loses electrons becomes positively charged, while the body which receives electrons becomes negatively charged.

When bodies of like charges are placed near to each other, they will ~~attract~~ repel each other while the bodies of unlike charges attract each other.

Coulomb's Law

According to Coulomb's law, the magnitude of force of attraction or repulsion b/w any two charged bodies depends upon.

- ① directly proportional to the product of their charges
- ② Inversely proportional to the square of the distance b/w them.
- ③ The nature of the medium b/w the charges.

The electrostatic force b/w two charges Q_1 & Q_2 placed at a distance ' r ' from each other,

$$F \propto Q_1, Q_2$$

$$F \propto \frac{1}{r^2}$$

$$F \propto \frac{Q_1 Q_2}{r^2}$$

$$= k \cdot \frac{Q_1 Q_2}{r^2}$$

where k is a constant of proportionality & depends upon the nature of the medium b/w two charges & is called the specific inductive capacity or the dielectric constant of the medium.

$$k = \frac{1}{4\pi \epsilon_0 \epsilon_r}$$

ϵ_0 is the permittivity of the vacuum

ϵ_r the relative permittivity of the medium b/w the two charges.

$$F = \frac{Q_1 Q_2}{4\pi \epsilon_0 \epsilon_r r^2}$$

Unit $F \rightarrow$ Newtons, (N)

$Q_1, Q_2 \rightarrow$ coulomb (C)

$r \rightarrow$ metre

$\epsilon \rightarrow$ permittivity (F/m)

$$\epsilon_0 = 8.854 \times 10^{-12} \text{ F/m}$$

for air $\epsilon_r = 1$;

$$F = \frac{Q_1 Q_2}{4\pi \epsilon_0 r^2}$$

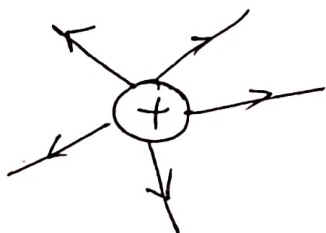
If $Q_1 = Q_2 = 1$ & $r = 1$;

$$F = \frac{1}{4\pi \epsilon_0} = \frac{1}{4\pi \times 8.854 \times 10^{-12}} \\ = \underline{\underline{8.99 \times 10^9 \text{ N}}}$$

A unit charge is defined as that charge, which when placed at a distance 1m in air from an equal & similar charge, will repel it with a force of $8.99 \times 10^9 \text{ N}$.

Electric field strength

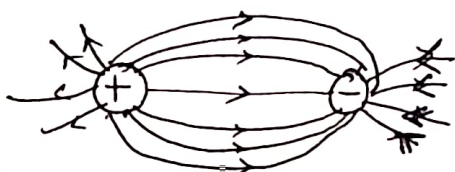
Electric field:- Any region of space, in which a charged body experiences a force of attraction or repulsion is called an electric field.



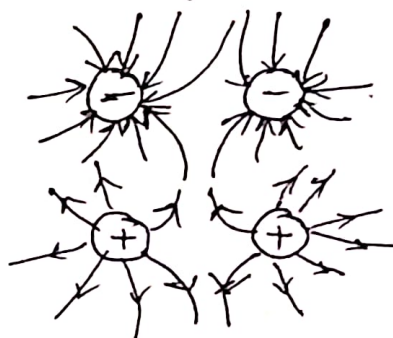
Lines of force for +ve charge



Lines of force for an isolated negative charge.



Lines of force due to two equal & opposite charges.



Lines of force due to two equal & similar charges.

Electric field strength (E)

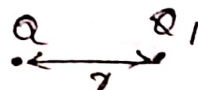
Force experienced by a unit positive charge placed at that point in the field is defined as the electric field strength or electric intensity of an electrostatic field at a point.

It is denoted by 'E'. It is a vector quantity. & its direction

is that in which an isolated positive charge is forced to move.

Unit is newton/Coulomb (N/C)

$$E = \frac{F}{Q_1} \text{ N/C}$$



$$F = \frac{Q_1 Q_2}{4\pi\epsilon_0 \epsilon_r r^2}$$

$$E = \frac{Q}{4\pi\epsilon_0 \epsilon_r r^2} \text{ (N/C)}$$

for a group of charges,

$$E = E_1 + E_2 + E_3 + \dots$$

Electric flux (ψ)

The total number of lines of force emanating from a certain charge is called electric flux.

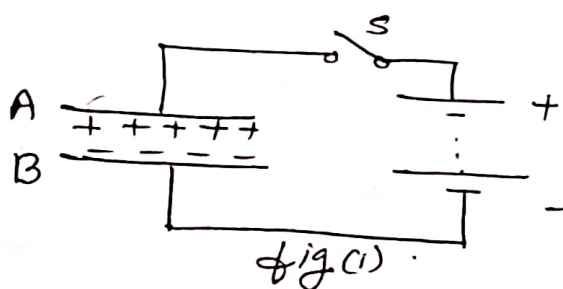
Electric flux density

The quantity ϵE has dimensions of charge/area or surface charge density. The product ϵE is electric flux density or displacement, designated by symbol ' D '. It is measured at right angles to the direction of electric field in Coulomb per square metre.

$$D = \epsilon E = \frac{Q}{4\pi r^2}.$$

Capacitors & capacitance.

Any two metal plates b/w which an electric field can be maintained constitutes a capacitor.



This arrangement has the capacity of storing electricity as an excess of electrons on one plate & a deficiency on the other.

fig (2) shows a ~~single~~ simple parallel plate capacitor. Let it be connected to a battery through leads of negligible resistance. When switch S is closed, there is a momentary flow of current. This flow of current ceases when plate A attains the same potential as the positive terminal of the battery & plate B the same potential as the -ve terminal. Now plate A of the capacitor is deficient in electrons

& thus it is positively charged. Plate B of the capacitor has an excess of electrons & carries a negative charge equal in amount to the positive charge on plate A. The capacitor is said to be in a charged state.

The charge Q on each plate of the capacitor is measured in coulomb. Charge Q on each plate is directly proportional to the p.d. b/w the two plates. The ratio of the charge Q on either plate of the capacitor to the p.d. b/w the plates is called the capacitance or the capacity of capacitor,

$$C = \frac{Q}{V}$$

The capacity of a capacitor depends upon the following factors :-

- (i) ~~area of the plate~~ :- is directly proportional to the surface area of the plate.
- (ii) is inversely proportional to the distance b/w the plates.
- (iii) is directly proportional to the permittivity of the intervening medium b/w the two plates.

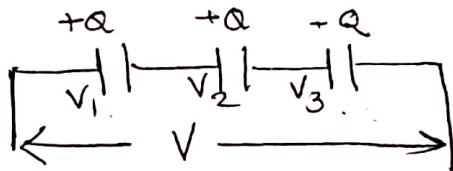
If three capacitors, C_1 , C_2 , & C_3 are connected in series, then the equivalent capacitance ' C '

$$\frac{1}{C} = \frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3}$$

If connected in parallel,

$$C = C_1 + C_2 + C_3$$

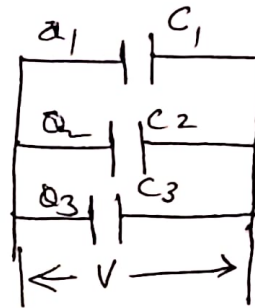
Energy stored = $\frac{1}{2} C V^2$.



$$V = V_1 + V_2 + V_3$$

$$\frac{Q}{C} = \frac{Q}{C_1} + \frac{Q}{C_2} + \frac{Q}{C_3}$$

$$\frac{1}{C} = \frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3}$$



$$Q = Q_1 + Q_2 + Q_3$$

$$C V = C_1 V + C_2 V + C_3 V$$

$$C = C_1 + C_2 + C_3$$

Types of Capacitors.

Mica

Usually, they are multiplate type consisting of a series of plates, alternate plates being connected together & separated by a dielectric. Generally, the plates are of metal foil separated by thin mica sheets. They have low pf on ac & hence are very suitable for use at radio frequencies.

Ceramic capacitors

These consists of disks of ceramic material whose opposite faces are coated with metallic silver. The ceramic discs acts as the dielectric & silver coatings as plate. They have low Pf which decreases with increase in freq. They are very useful for short wave work in radio.

Paper capacitors

These consists of sheets of metal foils usually Al or Cu or tin interleaved with thin paper impregnated with oil or wax. The foils and paper are then rolled into a compact form & are housed in sealed tubes of cardboard or bakelized paper or in oil-filled metal boxes. They are used in audio-freq stages of radio receivers as bypass & coupling capacitors.

Electrolytic capacitors

There are wet or dry type. The wet consists of a aluminium anode which is

centrally mounted in a metal cylinder filled with an electrolytic solution usually aluminium borate which acts as a cathode.

Energy stored in a capacitor.

When a capacitor is being charged, some work is done & this work done is stored by the capacitor as the energy of the charged capacitor. When the capacitor is discharged, this stored energy in the capacitor is released as heat.

Let the capacitance of the capacitor be C farad & the final ⁺ charge on one of its plates be Q coulomb with a p.d. V b/w the plates of Volts. The final charge Q on one of the plates of the capacitor may be supposed to be the result of bringing in succession small charges dq to the plates. Let q be the instantaneous charge on the positive plate as a result of bringing charge dq & V be the corresponding p.d. b/w the plates.

$$V = \frac{q}{C}$$

Work done in bringing a charge dq to the plate of the capacitor, $dw = V \times dq = \frac{q}{C} \times dq$

The total work done in charging the capacitor from zero to a charge $+Q$,

$$W = \int_0^Q \frac{q}{C} dq \quad \text{or} \quad W = \frac{1}{2} \frac{Q^2}{C}.$$

$$Q = CV.$$

$$W = \frac{1}{2} \frac{C^2 V^2}{C} = \frac{1}{2} C V^2$$

$$\text{Energy stored} = \underline{\underline{\frac{1}{2} C V^2}}.$$